our ultimate theory to its full sprawling expanse, and that our "extended ultimate theory" might tell us precisely why and how the values of the fundamental parameters are sprinkled across the constituent universes.

A more radical musing comes from a proposal of Lee Smolin of Penn State University, who, inspired by the similarity between conditions at the big bang and at the centers of black holes—each being characterized by a colossal density of crushed matter—has suggested that every black hole is the seed for a new universe that erupts into existence through a big bang-like explosion, but is forever hidden from our view by the black hole's event horizon. Beyond proposing another mechanism for generating a multiverse, Smolin has injected a new element—a cosmic version of genetic mutation—that does an end run around the scientific limitations associated with the anthropic principle.9 Imagine, he suggests, that when a universe sprouts from the core of a black hole, its physical attributes, such as particle masses and force strengths, are close, but not identical, to those of its parent universe. Since black holes arise from extinguished stars, and star formation depends upon the precise values of the particle masses and force strengths, the fecundity of any given universe—the number of black hole progeny it can produce—depends sensitively on these parameters. Small variations in the parameters of the progeny universes will therefore lead to some that are even more optimized for black hole production than their parent universe, and have an even greater number of offspring universes of their own.10 After many "generations," the descendants of universes optimized for producing black holes will thus be so numerous that they will overwhelm the population of the multiverse. And so, rather than invoking the anthropic principle, Smolin's suggestion provides a dynamic mechanism that, on average, drives the parameters of each next-generation universe ever closer to particular values—those that are optimum for black hole production.

This approach gives another method, even in the context of the multiverse, in which the fundamental matter and force parameters can be explained. If Smolin's theory is right, and if we are a typical member of a mature multiverse (these are big "ifs" and can be debated on many fronts, of course), the parameters of the particles and forces